Module 10: LAN Security Concepts

Switching, Routing and Wireless Essentials v7.0 (SRWE)

Module Objectives

Module Title: LAN Security Concepts

Module Objective: Explain how vulnerabilities compromise LAN security

Topic Title	Topic Objective
Endpoint Security	Explain how to use endpoint security to mitigate attacks
Access Control	Explain how AAA and 802.1x are used to authenticate LAN endpoints and devices
Layer 2 Security Threats	Identify Layer 2 vulnerabilities
MAC Address Table Attack	Explain how a MAC address table attack compromised LAN security
LAN Attacks	Explain how LAN attacks compromise LAN security

10.1 Endpoint Security

Network Attacks Today

- **Distributed Denial of Service (DDoS).**
- **Data Breach**
- **Malware**

- For example, ransomware such as WannaCry encrypts the data on a host and locks access to it until a ransom is paid.

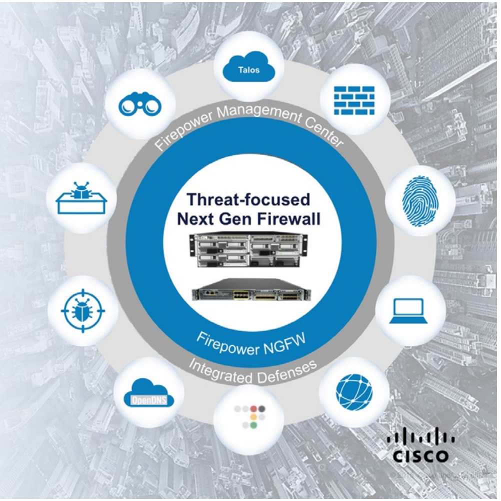
Network Security Devices

Various network security devices are required to protect the network perimeter from outside access. These devices could include the following:

- Virtual Private Network (VPN) enabled router - provides a secure connection to remote users across a public network and into the enterprise network. VPN services can be integrated into the firewall.

Network Security Devices

- Network Access Control (NAC) - includes authentication, authorization, and accounting (AAA) services. The Cisco Identity Services Engine (ISE) is an example
- Next-Generation Firewall (NGFW) - provides stateful packet inspection, application visibility and control, a next-generation intrusion prevention system (NGIPS), advanced malware protection (AMP), and URL filtering.



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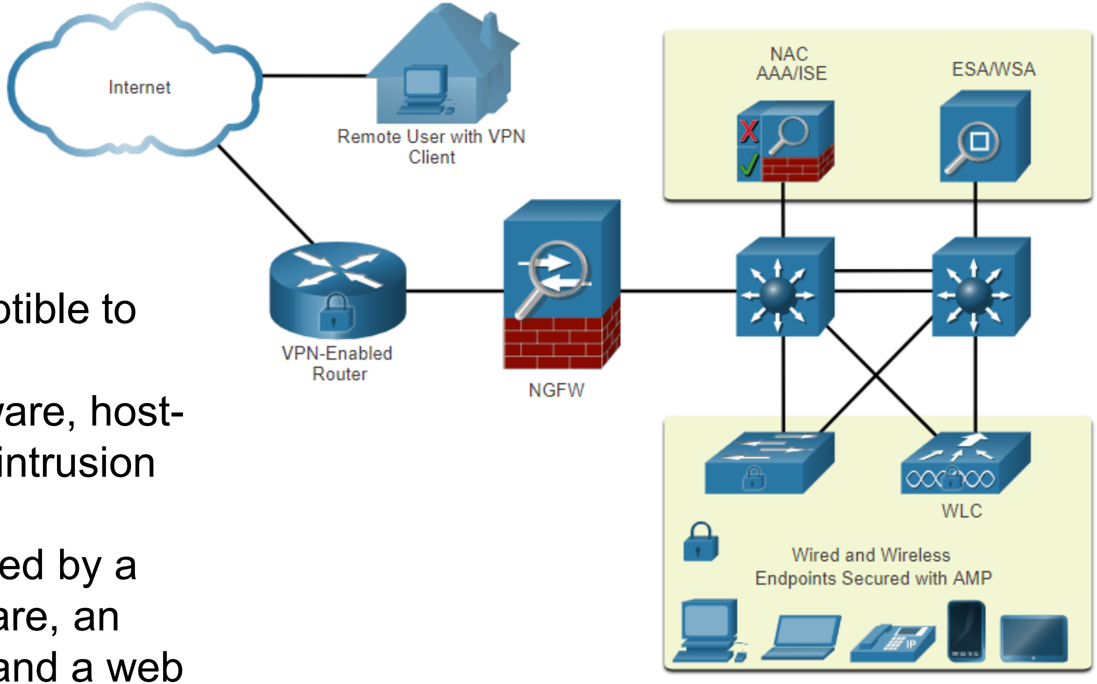
Endpoint Protection



- Endpoints are particularly susceptible to malware-related attacks
- Endpoints use antivirus/antimalware, host-based firewalls, and host-based intrusion prevention systems (HIPSs).
- Endpoints today are best protected by a combination of NAC, AMP software, an email security appliance (ESA), and a web security appliance (WSA).

Cisco Email Security Appliance

The Cisco ESA device is designed to monitor Simple Mail Transfer Protocol (SMTP)

Updated by real-time feeds from the Cisco Talos

This threat intelligence data is pulled by the Cisco ESA every three to five minutes.

These are some of the functions of the Cisco ESA:

- Block known threats
- Remediate against stealth malware that evaded initial detection
- Discard emails with bad links
- Block access to newly infected sites.
- Encrypt content in outgoing email to prevent data loss.

Cisco Web Security Appliance

- Mitigation technology for web-based threats.
- It helps organizations address the challenges of securing and controlling web traffic.
- Advanced malware protection
- Application visibility and control,
- Acceptable use policy controls,
- Incident reporting.
- complete control over how users access the internet. Certain features and applications, such as chat, messaging, video and audio, can be allowed, restricted with time and bandwidth limits, or blocked, according to the organization's requirements.
- The WSA can perform blacklisting of URLs, URL-filtering, malware scanning, URL categorization, Web application filtering, and encryption and decryption of web traffic.